

Activity - 2

Stack Queue & Linked List Questions

Answers

①

Stack:

push(54), push(52), pop()

→ 52 is removed

push(55), push(62), pop()

$S \neq 62$

Queue:

Enqueue(21), enqueue(24), dequeue()

→ 21 is removed

enqueue(28), enqueue(32).

$q = \text{dequeue} \rightarrow q = 24$

$S + q = 62 + 24 = 86$

② Postfix expression

$$105 + 606 / + 8 -$$

$$= (10 + 5) + (60 \div 6) - 8$$

$$= 15 + 10 - 8$$

$$= 17$$

③ Minimum Enqueue operations = 10

④ REVERSE does not change the computational power of stack

⑤ Postfix expression

$$823^* / 23^* + 51^* -$$

Top two elements are 6, 8

⑥ Infix to Postfix

$$a + b \times c - d^e f$$

$$abc^* + def^{**} -$$

⑦ push(10), push(20), pop, push(10),
push(20), pop, pop, push(20), pop

Sequence of popped values

Ans = 20, 20, 10, 20.